

Building Petro Refinery's Model

Module 6, Part 1 (December 5)

QBA 5330

Where We're Headed Today

- **Part 1:** Building Petro Refinery's model
- **Part 2:** The value of an extra hour – sensitivity analysis
- **Part 3:** Executive recommendation and course wrap-up

Petro Refinery LLC

A crude oil refinery deciding how much **light** and **heavy** crude to process each week.

- X_1 = barrels of light crude (thousands)
- X_2 = barrels of heavy crude (thousands)

The Model

Maximize $10X_1 + 12.5X_2$

Distillation: $2X_1 + 2X_2 \leq 3,000$

Conversion: $2X_1 + 3X_2 \leq 2,400$

Blending: $3X_1 + 2.5X_2 \leq 2,400$

Why Not Just Make Heavy Crude?

- Heavy crude's margin is higher (\$12.5 vs. \$10)
- But heavy crude also uses **more** conversion and blending per barrel
- **The optimal mix is a genuine trade-off, not a ranking.**

Live Build: Excel Solver

1. Decision variable cells + objective cell
2. One usage cell per constraint
3. Solver: Set Objective (Max), By Changing Cells, Subject to Constraints
4. Simplex LP, Assume Non-Negative
5. **Solve**

The Answer

300,000 barrels of light crude. 600,000 barrels of heavy crude.

\$10,500,000 per week in contribution.

Verifying in R

```
1 library(lpSolve)
2 obj <- c(10, 12.5)
3 mat <- matrix(c(2,2, 2,3, 3,2.5), nrow = 3, byrow = TRUE)
4 rhs <- c(3000, 2400, 2400)
5 sol <- lp("max", obj, mat, c("<=", "<=", "<="), rhs)
6 sol$solution
```

```
[1] 300 600
```

```
1 sol$objval
```

```
[1] 10500
```

Binding or Not?

```
1 used <- mat %*% sol$solution
2 data.frame(resource = c("Distillation", "Conversion", "Blending"),
3   available = rhs, used = as.numeric(used), slack = rhs - as.numeric(used))
```

	resource	available	used	slack
1	Distillation	3000	1800	1200
2	Conversion	2400	2400	0
3	Blending	2400	2400	0

- Conversion & blending: **fully used**
- Distillation: **1,200 hours of slack (only 60% used)**

Before the Break

Would buying more distillation capacity increase Petro's profit?

Next: putting a precise dollar value on every resource.